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Key Points:

- We systematically measure instrument response, noise levels, and amplitude saturation of telecommunication fiber-optic Distributed Acoustic Sensing arrays
- Our results suggest that the saturation amplitude of DAS is tunable for earthquake early warning for a magnitude 6+ earthquake at 10 km
- We find local ambient traffic noise levels affect the quality of interferometry results, crucial for near-surface imaging and monitoring

Supporting Information:

Supporting Information may be found in the online version of this article.

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Comprehensive Evaluation of DAS Amplitude and Its Implications for Earthquake Early Warning and Seismic Interferometry

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Abstract Distributed Acoustic Sensing (DAS) is an emerging technology that converts optical fibers into dense arrays of strainmeters, significantly enhancing our understanding of earthquake physics and Earth's structure. While most past DAS studies have focused primarily on seismic wave phase information, accurate measurements of true ground motion amplitudes are crucial for comprehensive future analyses. However, amplitudes in DAS recordings, especially for pre-existing telecommunication cables with uncertain fiber-ground coupling, have not been fully quantified. By calibrating three DAS arrays with co-located seismometers, we systematically evaluate DAS amplitudes. Our results indicate that the average DAS amplitude of earthquake signals closely matches that of co-located seismometer data across frequencies from 0.01 to 10 Hz. The noise floor of DAS is comparable to that of strong-motion stations but higher than that of broadband stations. The saturation amplitude of DAS is adjustable by modifying the pulse repetition rate and gauge length. We also demonstrate how our findings enhance the understanding of fiber-optic seismology and its implications for natural hazard mitigation and Earth structure imaging and monitoring. Specifically, our results suggest that with proper settings, DAS can detect *P*-waves from an M6+ earthquake occurring 10 km from the cable without saturation, indicating its viability for earthquake early warning. Through quantitative comparison and analysis, we also find that local ambient traffic noise levels strongly affect the quality of seismic interferometry measurement, which is a powerful tool for near-surface imaging and monitoring. Our methodology and findings are valuable for future DAS experiments that require precise seismic amplitude measurements.

Plain Language Summary Distributed Acoustic Sensing (DAS) is an innovative technology that transforms optical fiber into sensors capable of detecting ground vibrations, offering new insights into how earthquakes happen and the structure of the Earth. While previous studies using DAS mostly focused on the phase information in seismic waves, measuring the actual amplitude of ground motion is also important. However, the measured amplitudes in DAS recordings, especially for established communication cables, have yet to be quantitatively evaluated due to limited knowledge about how well the cables are coupled with the ground. We compared DAS with conventional seismic sensors at the same location and found that these DAS arrays can accurately capture earthquake signals. The minimum resolution of DAS is comparable to some low-cost conventional sensors but is not as good as the most sensitive seismometers. Importantly, we showed that DAS can be fine-tuned to detect large earthquakes without being saturated, making it a promising tool for earthquake early warning. Additionally, our results show how local traffic noise can impact the quality of DAS measurements, which is important for imaging and monitoring the shallow Earth. Our methodology and findings will help improve future DAS experiments that need accurate measurements of ground motion.

1. Introduction

Distributed Acoustic Sensing (DAS) is a rapidly developing technology that converts up to one hundred kilometers of optical fiber into a dense array of thousands of strainmeters, with sensor spacing at meter scale (Cheng, 2024; Parker et al., 2014; Waagaard et al., 2021). A DAS interrogator, by sending repeated laser pulses into the fiber and measuring phase shifts in Rayleigh backscattered light, can be used to infer axial strain or strain rates along a dedicated or telecommunication cable (Hartog, 2017; Lindsey & Martin, 2021; Zhan, 2019). Distributed Acoustic Sensing has considerably advanced seismological studies, enhancing our understanding of earthquakes and Earth's structure (Ajo-Franklin et al., 2019; Lior, Sladen, Mercerat, et al., 2021; Martin et al., 2017; Nayak et al., 2021; Spica et al., 2020; Zhai, Zhan, & Chavarria, 2024). It also shows potential for

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applications in planetary science (W. Wu et al., 2024; Zhai, Husker, et al., 2024). While most DAS research has focused on seismic wave phase information for investigating travel times and wavefield velocities, accurate measurements of true ground motion amplitudes are critical for other comprehensive seismological analyses, such as full-waveform inversion (Egorov et al., 2018), attenuation (Sayed et al., 2021), and moment tensor inversion (Lecoulant et al., 2023). However, in contrast to conventional permanent seismic stations, DAS's amplitude is challenging to interpret due to the uncertain fiber-ground coupling (Ide et al., 2021), particularly with pre-existing telecommunication cables (Lior, Sladen, Rivet, et al., 2021). Despite these challenges, the extensive networks of on-land and offshore telecommunication cables offer a unique and cost-effective opportunity for the development of regional and global DAS networks (Lindsey et al., 2019; Spica et al., 2023; Wuestefeld et al., 2023). Therefore, gaining a comprehensive understanding of DAS amplitude information is crucial for the future development of DAS as a measurement tool.

Three questions are fundamental to a comprehensive understanding of DAS amplitude: (a) What is the transfer function between true ground motion and observed DAS data? Understanding this transfer function (i.e., instrument response) is important for accurately resolving true ground motion. (b) What is the minimum amplitude DAS can measure? Identifying this noise floor is crucial for detecting weak signals of small or remote earthquakes. (c) What is the maximum amplitude it can record? The maximum (i.e., saturation/clipping) amplitude (Viens et al., 2022), is essential for analyzing large earthquakes' source properties and implementing early warning systems (Ende et al., 2024; Lior et al., 2023; Yin, Soto, et al., 2023; Yin, Zhu, et al., 2023). Distributed Acoustic Sensing measures optical phase changes that undergo cycle skipping. As the phase approaches its limits ($-\pi$ or π), it skips to the next cycle, making the exact point of saturation unclear.

Although recent studies have advanced our understanding of these questions, they have not been comprehensively answered due to various limitations. Notably, only the instrument response of Silixa's iDAS (Parker et al., 2014) has been well documented (Jousset et al., 2018; Lindsey et al., 2020; Muir & Zhan, 2021; Païtz et al., 2021), despite the presence of numerous DAS manufacturers. Additionally, much of the research has been limited to data from a single DAS cable, showing inconsistent findings across studies. While the signal-to-noise ratio (SNR) and power spectral density (PSD) of DAS recordings have been extensively explored (Cox et al., 2012; Daley et al., 2013, 2016; Ide et al., 2021; Lior, Sladen, Rivet, et al., 2021; Matsumoto et al., 2021; Parker et al., 2014; Shinohara et al., 2022; Wang et al., 2018), research into DAS's noise level and seismicity monitoring capacity is immature compared to that of conventional seismometers (Clinton & Heaton, 2002; Ringler & Hutt, 2010). For example, we do not know the minimum magnitude of an earthquake at a given distance a DAS array can detect. Additionally, the issue of amplitude saturation/clipping has been reported (Abukrat et al., 2023; Diaz-Meza et al., 2023; Viens et al., 2022), but requires further documentation. For instance, we do not know the maximum earthquake magnitude that can be recorded without signal saturation at a given distance or how can we quantitatively adjust parameters in DAS settings to achieve the desired dynamic range.

In this study, we calibrated three telecommunication fiber-optic DAS arrays in California using co-located seismometers to systematically evaluate their performance. In the following sections, we first describe the three DAS arrays under examination. We then explore three critical aspects of DAS amplitude: (a) instrument response, (b) noise levels, and (c) amplitude saturation. In the discussion section, we explain how our new findings advance our understanding of fiber-optic seismology and its implications for natural hazard mitigation and Earth structure imaging and monitoring. We specifically address the potential of DAS for earthquake early warning (EEW) (Lior et al., 2023; Yin, Soto, et al., 2023), with a focus on the challenge of amplitude saturation in estimating the magnitude of large earthquakes. Additionally, we examine how local ambient (traffic) noise levels affect the quality of DAS seismic interferometry results, which is crucial for near-surface imaging and monitoring (Ajo-Franklin et al., 2019; Dou et al., 2017; Z. Shen et al., 2024; Spica et al., 2020; Yang, Atterholt, et al., 2022; Yang, Zhan, et al., 2022). Finally, we summarize our findings and highlight their implications for future research.

2. Data

California provides an ideal setting for assessing the performance of telecommunication fiber-optic DAS arrays and their calibration with conventional seismometers for several reasons. First, the region is covered by extensive telecommunication fiber networks. Second, it has been well monitored by dense seismic networks such as the Southern California Seismic Network (SCSN) (SCEDC, 2013). Additionally, the high seismic activity in California offers abundant earthquake signals for detailed analysis and calibration.

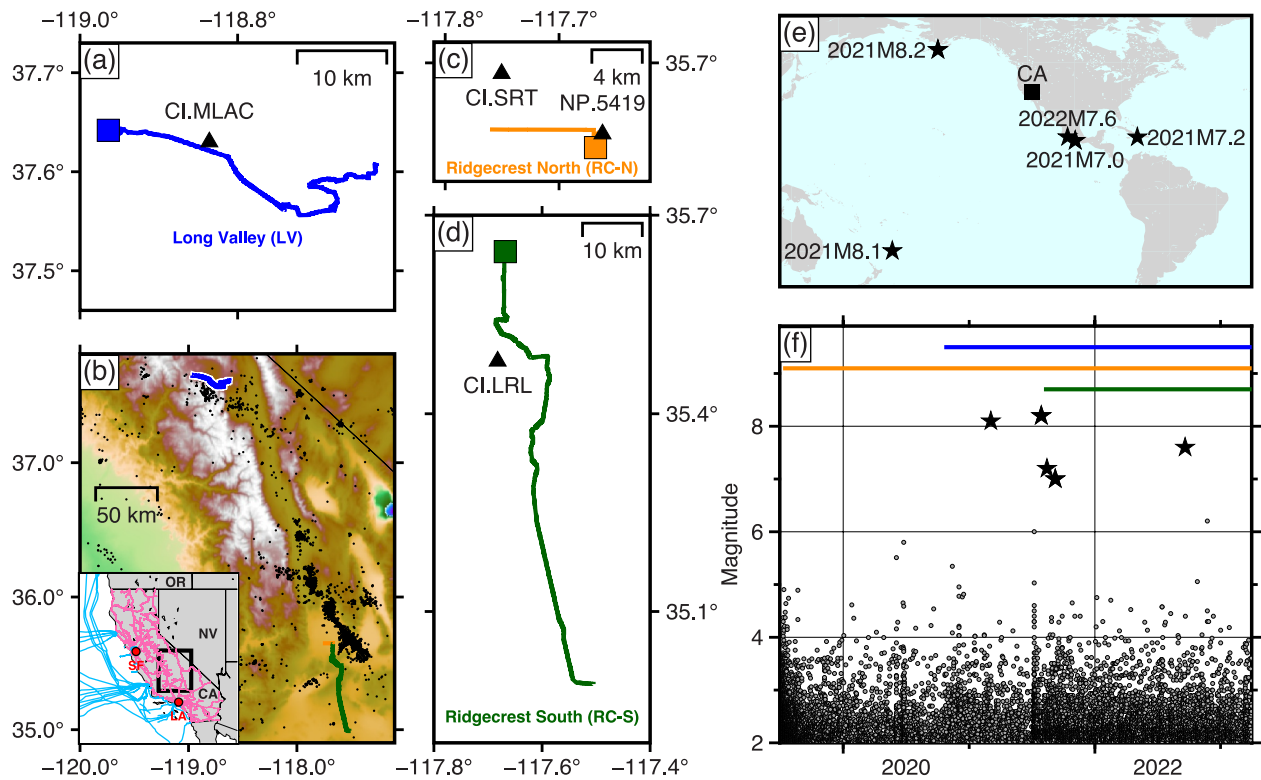


Figure 1. Map of three Distributed Acoustic Sensing (DAS) arrays in California. (a) Long Valley (LV, blue curve) DAS array as an enlarged view of the section from (b), (c) Ridgecrest North (RC-N, orange curve) DAS array, (d) Ridgecrest South (RC-S, green curve) DAS array. These three DAS arrays are small portions of California's telecommunication fiber networks, illustrated by the pink and blue lines in the inset map of (b). Colored squares in (a), (c), and (d) indicate DAS interrogators at one end of the cables. Black triangles in (a), (c), and (d) mark co-located permanent seismic stations from the Southern California Seismic Network, (e) Map of teleseismic events (black stars) used in this study. The black square in (e) indicates the study area in California, as shown in (b), and (f) The magnitude-time distribution of teleseismic events (black stars) and local events (black dots, same as they are in (b)) used in this study. The colored lines in (f) denote the operational periods of the three DAS arrays with colors corresponding to those used in (a)–(d).

This study examines three DAS arrays (Figure 1), each connected to an OptaSense DAS instrument within segments of California's telecommunication fiber networks: Long Valley (LV, Figure 1a), Ridgecrest North (RC-N, Figure 1c), and Ridgecrest South (RC-S, Figure 1d). These DAS arrays have been extensively used to study earthquake source properties and Earth's structure (Atterholt et al., 2021, 2022, 2024; Biondi et al., 2022, 2023; J. Li, Kim, et al., 2023; J. Li, Zhu, et al., 2023; Z. Li et al., 2021; Z. Shen et al., 2024; Yang, Atterholt, et al., 2022; Yang, Zhan, et al., 2022; Yin, Zhu, et al., 2023; Zhu et al., 2023). These arrays span approximately 10–100 km with channel spacings of roughly 10 m (Table 1). Each array has continuous recordings for at least 1 year between 2019 and 2023 (Figure 1f), under various recording parameters and configurations, as detailed in Table 1. This variability enables the evaluation of how different configurations affect the performance of DAS instruments. All arrays are co-located with at least one permanent broadband seismic station from the SCSN (Figure 1). During our experiments, numerous local earthquakes cataloged by SCSN (Figures 1b and 1f), including aftershocks of the 2019 M7.1 Ridgecrest earthquake, were recorded by both the DAS arrays and co-located seismometers. In addition to these small local events, we also analyzed five high-quality ($M > 7$)

Table 1
Instrumental Configuration Parameters of Each Distributed Acoustic Sensing Array

	DAS interrogator	Cable length (km)	Channel number (#)	Channel spacing (m)	Gauge length (m)	Pulse repetition rate (Hz)
Long Valley	OptaSense QuantX	54	5,000	10.87	20.42	1,000
Ridgecrest North	OptaSense ODH3	10	1,250	8	16.34	5,000
Ridgecrest South	OptaSense QuantX	102	10,000	10.2	102.09	1,000

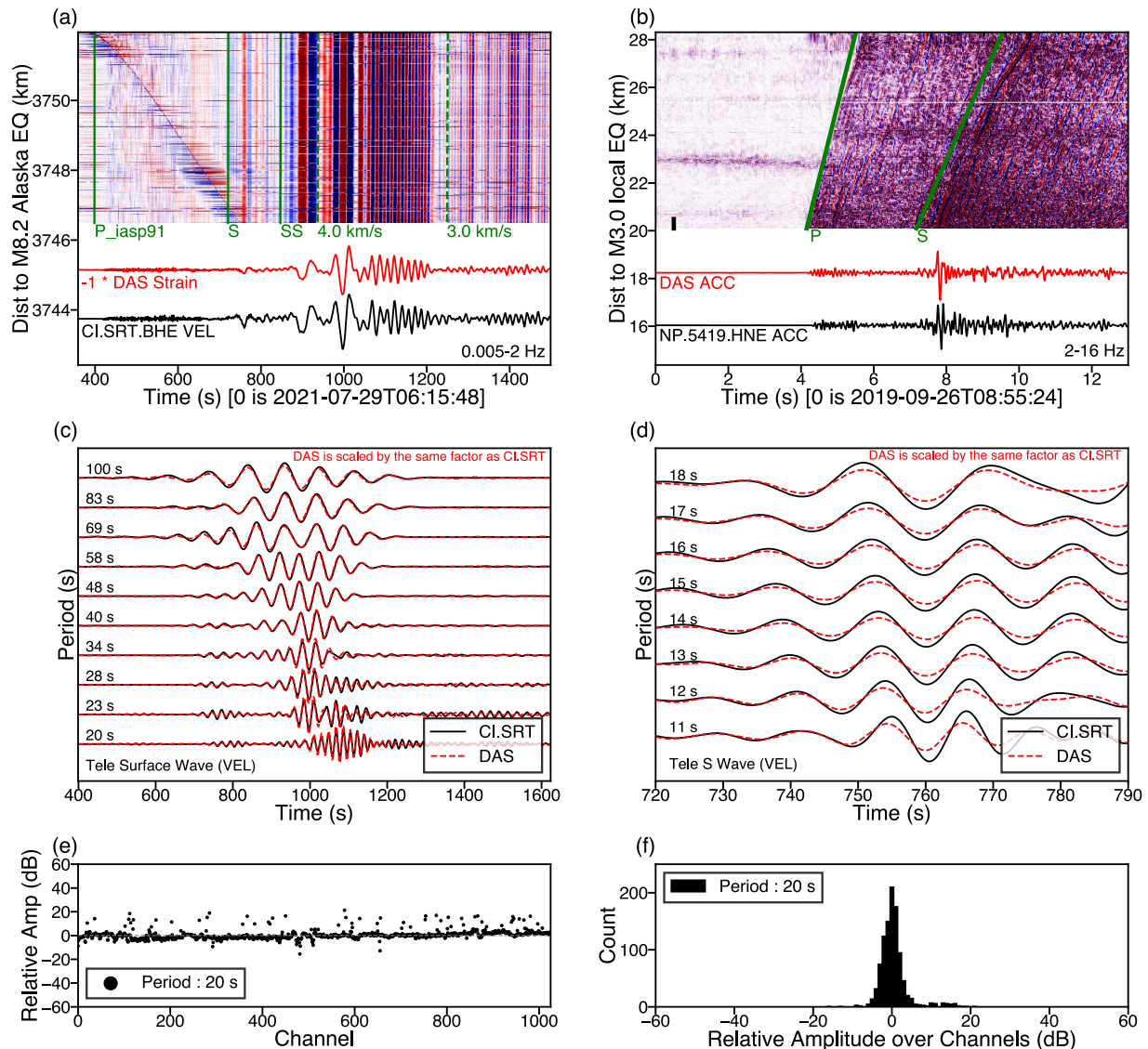


Figure 2. Comparison of Distributed Acoustic Sensing (DAS) waveforms with co-located seismometers and across different DAS channels. (a) Example recordings from the Ridgecrest North DAS array (as depicted in Figure 1c) of an M8.2 teleseismic event. Phase arrivals, calculated using the IASP91 velocity model (Kennett & Engdahl, 1991) or specified surface wave velocities, are indicated by green lines. The channel-average DAS strain, displayed in red with polarity inverted, is shown alongside the velocity recordings from the co-located CI.SRT broadband station (black waveform; refer to Figure 1c). Waveforms are normalized individually for enhanced visualization, (b) A comparative analysis between Ridgecrest North DAS and the co-located NP.5419 strong-motion accelerograph for an M3.0 local event, similar to panel (a). Phase arrivals are computed using a 1D local velocity model (Text S1 in Supporting Information S1). The DAS data, converted from strain rate to acceleration via Fast Discrete Curvelet Transform (Atterholt et al., 2021; Yang, Atterholt, et al., 2022), is represented by the red waveform, averaged over 50 channels closest to the NP.5419 station (indicated by the vertical black bar on the left), (c) This panel presents narrow-band filtered comparisons of surface wave recordings from Ridgecrest North DAS and the co-located seismometer. Red and black waveform pairs are normalized with an identical scaling factor to accurately reflect the true relative amplitude ratio, with the original broadband filter data shown in (a), (d) Similar to (c) but focusing on teleseismic S wave comparisons, (e), and (f) The relative amplitude distribution of the surface wave from the M8.2 teleseismic event across all channels of the Ridgecrest North DAS, as illustrated in (a) and (c).

teleseismic earthquakes (Figures 1e and 1, and Table S1 in Supporting Information S1) to compare DAS and seismometer signals at frequencies below 1 Hz.

3. Instrument Response

We empirically quantified the instrument response of the DAS arrays by calibrating them with co-located seismometers (Figure 1). To facilitate a comparison of ground motion as recorded by both DAS and

seismometers (Figure 2), we convert their recordings to the same physical units using the following equation (Lindsey et al., 2020; Paitz et al., 2021; Yu et al., 2019):

$$\begin{aligned} \dot{u}(\omega) &= -\varepsilon(\omega) \cdot c(\omega) \\ \text{or, } \ddot{u}(\omega) &= -\dot{\varepsilon}(\omega) \cdot c(\omega) \end{aligned} \quad (1)$$

where ω is the angular frequency. Here, $\dot{u}$ and $\ddot{u}$ denote the velocity and acceleration along the DAS cable as measured by the seismometer, respectively, while ε and $\dot{\varepsilon}$ represent the strain and strain rate recorded by DAS. The term c is the apparent velocity along the DAS cable. The DAS instrument amplitude response, denoted as $i(\omega)$, is defined as:

$$i(\omega) = \left| \frac{-\varepsilon(\omega) \cdot c(\omega)}{\dot{u}(\omega)} \right| = \left| \frac{-\dot{\varepsilon}(\omega) \cdot c(\omega)}{\ddot{u}(\omega)} \right| \quad (2)$$

Theoretically, $i(\omega)$ is expected to be equal to one down to zero frequencies (Lindsey et al., 2020). In practice, deviations from this ideal may occur considering variations in cable-ground coupling.

To empirically quantify $i(\omega)$, we take the following steps. (a) We select different types of target signals for different periods (T), teleseismic surface waves for T between 20 and 100 s, teleseismic S waves for T between 10 and 20 s, teleseismic P waves for T between 1 and 10 s, and body waves (P and S) of local earthquakes for T between 0.1 and 1 s. (b) We cut both the DAS and co-located seismometer recordings to the same time windows around the target signals (Figures 2a and 2b). We preprocess them by removing the mean and linear trend and tapering the waveform. To maximize compliance with the co-located assumption and avoid single-channel bias, the 50 DAS channels closest to the seismometer are used for the subsequent calibration. We also rotate the seismometer data to align with the direction along the closest DAS segment. (c) We convert the preprocessed broadband-station data to velocity using the instrument response information from SCSN for long-period ($T \geq 1$ s) teleseismic waveform. We convert short-period ($T < 1$ s) local event data recorded by strong-motion accelerometer to acceleration. These data are then narrow-band filtered into different periods (Figures 2c and 2d). (d) We filter the preprocessed teleseismic DAS data into the same periods using the identical filter and convert the filtered DAS data (Figure 2a) from strain to velocity by multiplying by the apparent velocity $c(\omega)$ along the DAS cable, determined using 1D velocity models (Figure S1 in Supporting Information S1). The 1D models include local-specific velocity models (Biondi et al., 2023; W. S. Shen et al., 2013; White et al., 2021; Yang, Atterholt, et al., 2022) for shallow depths and a global velocity model (Kennett & Engdahl, 1991) for the deeper depths (Text S1 in Supporting Information S1). (e) We convert the DAS data of local earthquakes (Figure 2b) from strain rate to acceleration using the Cartesian corone framework of the Fast Discrete Curvelet Transform (FDCT) (Atterholt et al., 2021; Candès et al., 2006) (Text S2 in Supporting Information S1), same as the method used in Yang, Zhan, et al. (2022). We then filter the DAS data into the same periods using the same filters. (f) For each period of a type of seismic signal (e.g., the $T = 100$ s teleseismic surface wave as shown at the top of Figure 2c), we measure the peak particle motion (velocity or acceleration) of both the DAS data (red dashed line in Figure 2c) and co-located seismometer recording (black line in Figure 2c). We then calculate the DAS instrument amplitude response, $i(\omega)$, by computing the DAS-to-seismometer peak particle motion ratio (Figure 3). Note that the DAS (red) and co-located seismometer (black) waveform pairs shown in Figures 2c and 2d are normalized using an identical scaling factor to accurately reflect the true relative amplitude ratio between them. Figure 2c presents the results based on the 50 DAS channels closest to the seismometer. Figure S2 in Supporting Information S1 provides single-channel and multi-channel example results.

The result of the collective average DAS instrument amplitude response, as illustrated in (Figure 3), is derived from all data across the three California telecommunication fiber-optic DAS arrays and multiple seismic phases (Figures 2a and 2b) from all analyzed earthquakes (Figure 1). Our results show that the average DAS instrument amplitude response between periods 0.1–100 s is approximately 0 dB (Figure 3), indicating that the observed DAS-to-seismometer signal amplitude ratio is about 1:1 (Figures 2c and 2d). The measurement error remains within a few dB across periods (Figure 3). The temporal variability over 1 month of continuous recording is shown in Figure S3 in Supporting Information S1. The spatial variability across channels is shown in Figures 2e and 2f, and Figure S4 in Supporting Information S1. Additionally, the wiggle-by-wiggle consistency between the

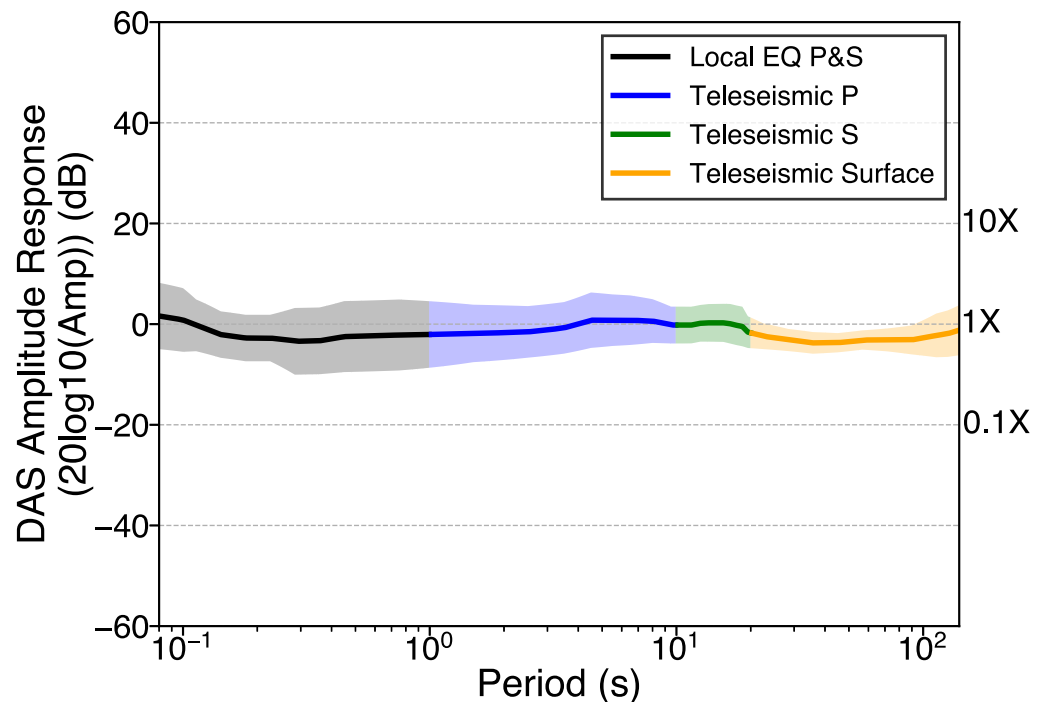


Figure 3. The average Distributed Acoustic Sensing instrument amplitude response. Data in different colors represent results from different types of waveforms. The shaded area delineates the standard deviation.

DAS data and seismometer data, as depicted in Figures 2c and 2d suggests that the phase shift between the true ground motion and DAS recordings is small, consistent with Lindsey et al. (2020) and Paitz et al. (2021).

4. Noise Level

We assessed the noise level of DAS arrays and compared them with co-located strong-motion and broadband stations (Figure 1). The DAS noise level is characterized by computing the noise power spectral density (PSD) of the DAS strain rate using the Fast Fourier Transform (FFT). The PSD for each channel of the three DAS arrays is calculated in 15 min segments. To mitigate the effects of temporal variations in background noise, we derive the median PSD from continuous recordings over 1 month. Similarly, we calculated the PSDs of acceleration for all co-located seismometers using the same duration and methodology. To facilitate a comparison between the DAS PSDs of strain rate and the seismometer PSDs of acceleration, we converted the channel-average PSD from strain rate to acceleration. This conversion uses Equation 1 and assumes that the background noise energy recorded by DAS mostly consisted of surface waves propagating along the DAS cable (Xiao et al., 2022), with velocities determined from the same local 1D velocity models used in the instrument response analysis.

Our results show that the noise floor of the DAS arrays is similar to that of the co-located strong-motion station over periods ranging from 0.1 to 100 s (frequencies from 0.01 to 10 Hz), but significantly exceeds that of the broadband stations periods longer than approximately 7 s (frequencies below approximately 0.14 Hz) (Figures 4a, 4c). We also performed a quantitative comparison of the DAS PSD against the Peterson New Low-/High-Noise Model as reported by Peterson (1993) and earthquake signals with various magnitudes at specific distances as measured by Clinton and Heaton (2002) (Figures 4 and 5). The DAS PSD exceeds the Peterson New High-Noise Model for periods longer than about 7 s (Figures 4 and 5), predominantly due to self-instrumental noise at low frequencies, similar to observations of the strong-motion station (Figure 4a). The strong noise in DAS at low frequencies (Figure S5 in Supporting Information S1) is attributed to the laser noise (Waagaard et al., 2021). Figures 4d and 4e show relative variations in the noise level across different channels along the Ridgecrest North DAS cable.

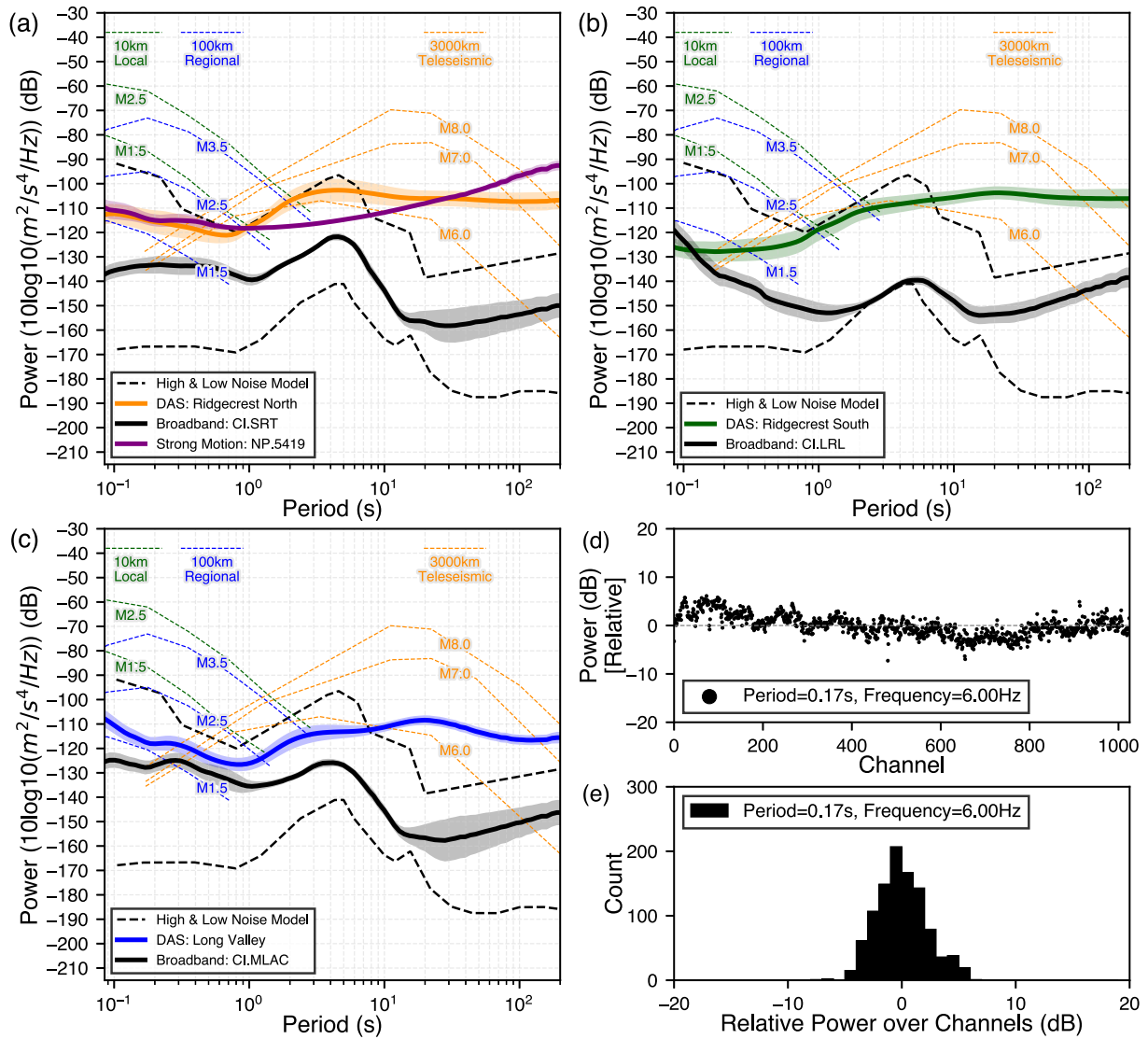


Figure 4. Comparison of Distributed Acoustic Sensing (DAS) noise level with co-located seismometers and across different DAS channels. (a) The orange solid line represents the power spectral density (PSD) of the Ridgecrest North DAS array. The orange shaded areas indicate the standard deviation of the measurements resulting from the temporal variability within 1 month of continuous recordings. The two black dashed lines show the Peterson New Low-/High-Noise Model (Peterson, 1993). The colored dashed lines indicate earthquake signals of various magnitudes at specified distances as measured by Clinton and Heaton (2002). The original earthquake signal data measured by Clinton and Heaton (2002) has a different unit compared to noise PSD. Therefore, to compare them on the same plot, the data must be converted to the same unit as PSD. This conversion has already been performed in ObsPy (Krischer et al., 2015) and is accessible at: https://docs.obspy.org/_modules/obspy/signal/spectral_estimation.html. (b)–(c) Similar to (a), but for the Ridgecrest South and Long Valley DAS arrays and their respective co-located seismometers (solid black lines), (d), and (e) The relative PSD distribution across all channels of the Ridgecrest North DAS.

5. Amplitude Saturation

A DAS interrogator sends repeated laser pulses into the fiber and measures the optical phase changes in Rayleigh backscatter. The phase change rate ($\dot{\phi}$) can be converted to axial strain rates ($\dot{\epsilon}$) along the fiber using the following equation (Lin et al., 2024; Yin, Zhu, et al., 2023):

$$\dot{\epsilon} = \frac{\lambda}{4\pi n \xi G} \dot{\phi} \quad (3)$$

where $\lambda = 1.55 \times 10^{-6}$ m is the laser wavelength, $n \approx 1.4682$ is the refractive index of the fiber, $\xi = 0.78$ is the photo-elastic scaling factor for longitudinal strain, and G is the gauge length. If the actual phase change between

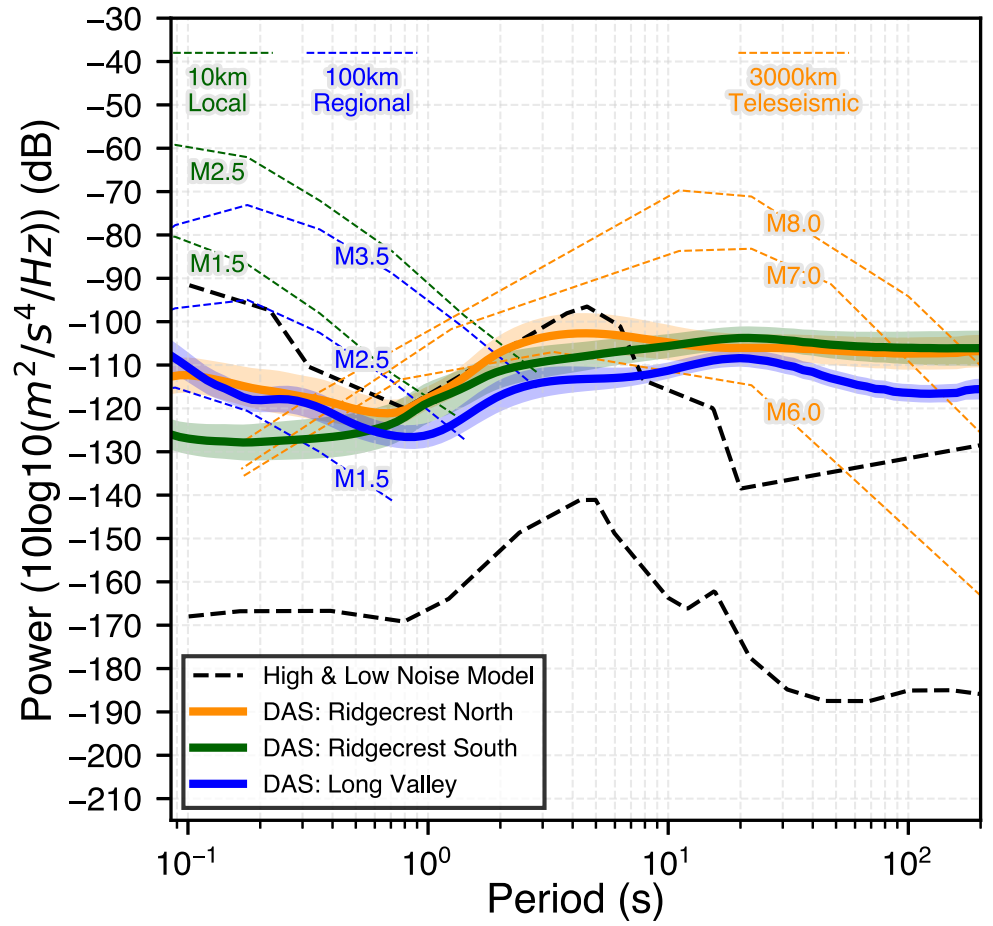


Figure 5. Comparison of the noise level of three Distributed Acoustic Sensing (DAS) arrays. The colored solid lines are the power spectral density of DAS arrays, the same as those in Figures 4a–4c. The shaded areas around these lines indicate the standard deviation, the same as those in Figure 4. The two black dashed lines show the Peterson New Low-/High-Noise Model (Peterson, 1993). The colored dashed lines indicate earthquake signals of various magnitudes at specified distances as measured by Clinton and Heaton (2002).

two pulses exceeds $-\pi$ or π , the recorded phase change ($\Delta\phi$) remains within the interval $[-\pi, \pi]$ due to the cycle skipping. This behavior differs from conventional seismometers, which do not experience cycle skipping. The theoretical maximum measurable strain rate (i.e., saturation/clipping) of DAS can be calculated as:

$$\begin{aligned}\dot{\epsilon}_{\text{saturation}} &= \frac{\lambda}{4\pi n \xi G} \max(|\dot{\phi}|) \\ &= \frac{\lambda}{4\pi n \xi G} \max_{-\pi < \Delta\phi \leq \pi} \left(\frac{|\Delta\phi|}{\Delta t_R} \right) \\ &= \frac{\lambda}{4n \xi G} f_R\end{aligned}\quad (4)$$

where Δt_R is the time interval between two pulses, and $f_R = 1/\Delta t_R$ is the pulse repetition rate.

We validate the theoretical saturation strain rate (Equation 4) using observations from three California DAS arrays. We first extract the DAS strain-rate waveforms of local earthquakes in the SCSN catalog (Figure 1). After preprocessing the data by removing the mean and linear trend, we measure the peak absolute strain rate for each earthquake on each channel. Figure S6 in Supporting Information S1 shows the relationship between earthquake magnitude and observed peak absolute strain rate. We then calculate the maximum peak strain rate of all earthquakes for each channel along each DAS array (Figure 6a). Our results show that the observed saturation

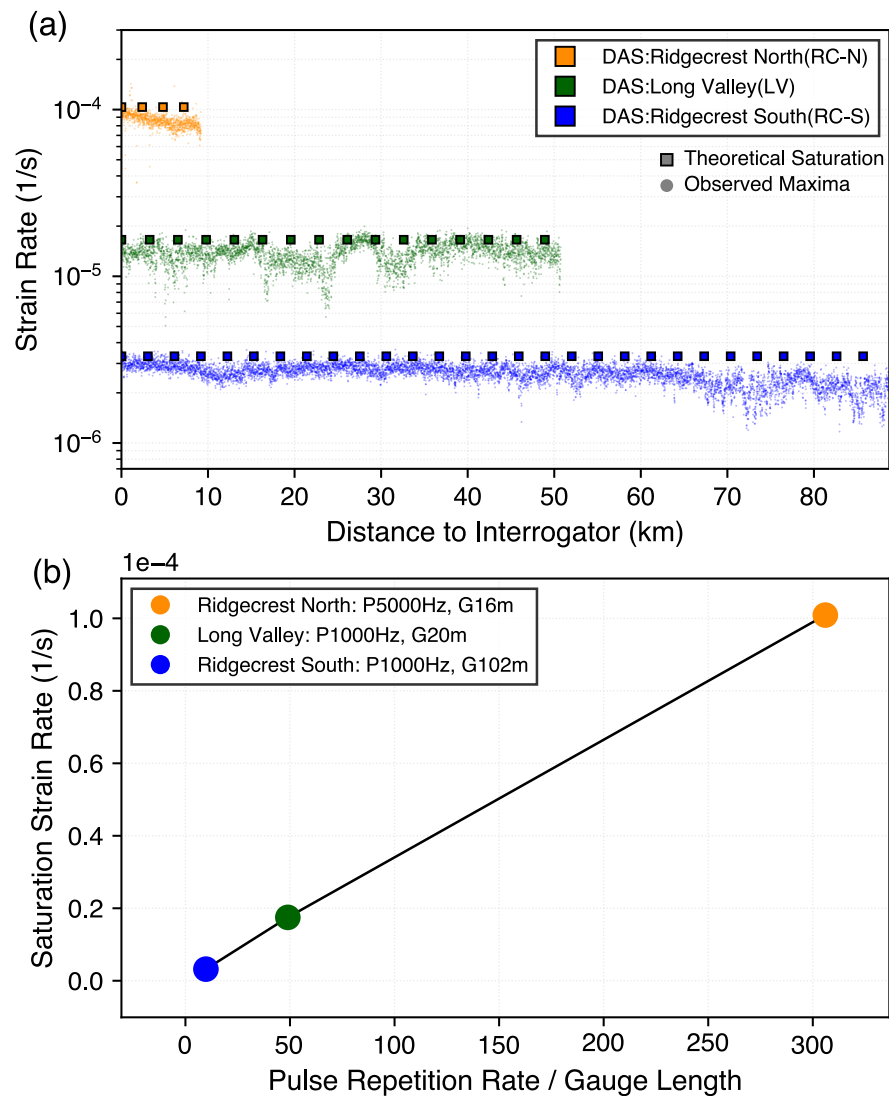


Figure 6. Amplitude saturation/clipping of Distributed Acoustic Sensing (DAS). (a) Colored squares represent the theoretical saturation strain rates for three DAS arrays, calculated using Equation 4. Small colored dots are the observed maximum peak strain rates on different DAS channels from all local earthquakes analyzed in this study (Figure 1). (b) The distribution of observed saturation strain rates alongside the pulse-repetition-rate (P) to gauge-length (G) ratio for DAS arrays (Table 1). The saturation strain rate for each DAS is determined by selecting the 98% highest values from all channels as shown in (a).

strain rates closely match the theoretical predictions (Figure 6a). Figure 6b illustrates that the observed saturation strain rate of DAS is proportional to the pulse-repetition-rate-to-gauge-length ratio, as suggested by Equation 4. Figure 6a and Figure S7 in Supporting Information S1 show the variations in the observed saturation strain rates across different channels.

6. Discussion

Using data from three DAS arrays and the co-located seismometers, we systematically measured the instrument response, noise level, and amplitude saturation of DAS. Figure 2c demonstrates that earthquake waveforms recorded by a DAS closely match those recorded by a co-located conventional seismometer, aligning wiggle by wiggle in both phase and amplitude. This suggests that DAS data can capture the same physical information about the earthquake source and Earth's structure. Consequently, scientific questions traditionally studied using conventional seismometer waveforms (e.g., those shown in Figure 2c) could also be investigated using DAS data, provided that potential errors in DAS recordings (e.g., absolute amplitude) are well-calibrated with co-located

conventional seismometers. Our results indicate that the cable-ground coupling of the three telecommunication cables used in this study is generally good (Figure 3). However, we emphasize that such good coupling may not be representative of all DAS field experiments, particularly for fibers that are not buried and are instead laid on the surface—whether on land, glaciers, or the ocean bottom (Ide et al., 2021; Lior, Sladen, Rivet, et al., 2021). Further investigation is needed to better understand the potential impacts of unburied fibers on DAS amplitude, which likely vary between DAS experiments and may also differ within each DAS array (Lindsey et al., 2020). Additionally, unburied cables may be more sensitive to noise from wind, which can reduce the signal-to-noise ratio (SNR) of DAS data (Donahue et al., 2024). The framework and methodology provided in this study can serve as a valuable tool for assessing the performance of other current and future DAS experiments.

In the following subsections, we demonstrate in detail how our new findings contribute to our understanding of fiber-optic seismology and its implications for natural hazard mitigation and Earth structure imaging and monitoring. Specifically, we thoroughly discuss EEW with DAS, addressing its amplitude saturation issue for magnitude estimation of large earthquakes. Additionally, we examine how local ambient (traffic) noise levels affect the quality of DAS seismic interferometry results, which is critical for near surface imaging and monitoring.

6.1. Earthquake Early Warning

Seismic hazard caused by large earthquakes is among the most devastating natural hazards. To mitigate it, EEW systems have been built in different regions around the world (Allen & Melgar, 2019). The key physical principle underlying EEW is that electromagnetic waves travel faster than seismic waves. Therefore, seismic sensors located close to a devastating earthquake are crucial for the success of an EEW system. The extensive pre-existing networks of on-land and offshore telecommunication cables offer a cost-effective opportunity for EEW, especially in subduction zone regions where deploying long-term real-time ocean bottom seismic networks is challenging and expensive.

Compared to the conventional strong-motion sensors that EEW systems mostly rely on, conventional broadband seismic stations have higher sensitivity (lower noise levels). However, their measurements are more likely to become saturated during the strong ground motion caused by local large earthquakes. The saturation of a conventional seismic sensor occurs when the feedback electronics can no longer keep the mass centered, a mechanism inherently designed within the device. In contrast, DAS saturation depends on cycle skipping and specific settings that can be adjusted. As an emerging seismic sensing technique, DAS also needs to have its possible saturation issues comprehensively assessed for EEW. If the DAS amplitude is saturated and directly used to calculate the earthquake magnitude for EEW, the earthquake magnitude and the associated seismic hazard will likely be underestimated. This possible issue has not been well addressed by previous studies that focus on assessing the feasibility of DAS for EEW using earthquake examples that do not saturate on DAS (Lior et al., 2023; Yin, Soto, et al., 2023).

In this study, we discuss and address the possible amplitude saturation issue of DAS with the following steps. First, we calculate empirical thresholds of DAS saturation as a function of earthquake magnitude and distance for DAS arrays with different pulse-repetition-rate to gauge-length ratios and amplitude saturation values. Next, we present a local M4.5+ earthquake example that does not saturate for either *P* or *S* waves with appropriate DAS settings. Then, we show an earthquake example with saturated *S* waves and unsaturated *P* waves. Our results suggest that saturated *S* waves underestimate the earthquake magnitude, while unsaturated *P* waves do not and can be used for EEW. Finally, we discuss and summarize strategies to solve or mitigate the possible amplitude saturation issue of DAS for EEW.

To get empirical DAS saturation thresholds as a function of earthquake magnitude and distance, we use the scaling relation of DAS data reported by Yin, Zhu, et al. (2023):

$$\log_{10} \dot{\epsilon}_{peak} = aM + b \log_{10} D + K \quad (5)$$

where $\dot{\epsilon}_{peak}$ is the peak DAS strain rate in microstrain/s (10^{-6} /s), M is the earthquake magnitude, D is the distance in kilometers, and a , b , and K are coefficients derived from fitting the observations of California DAS arrays. By setting the $\dot{\epsilon}_{peak}$ in Equation 5 equal to the observed $\dot{\epsilon}_{saturation}$ (Figure 6), we can determine the saturation thresholds for earthquake magnitude and distance. Due to the distinct scaling relationships for *P* and *S* waves

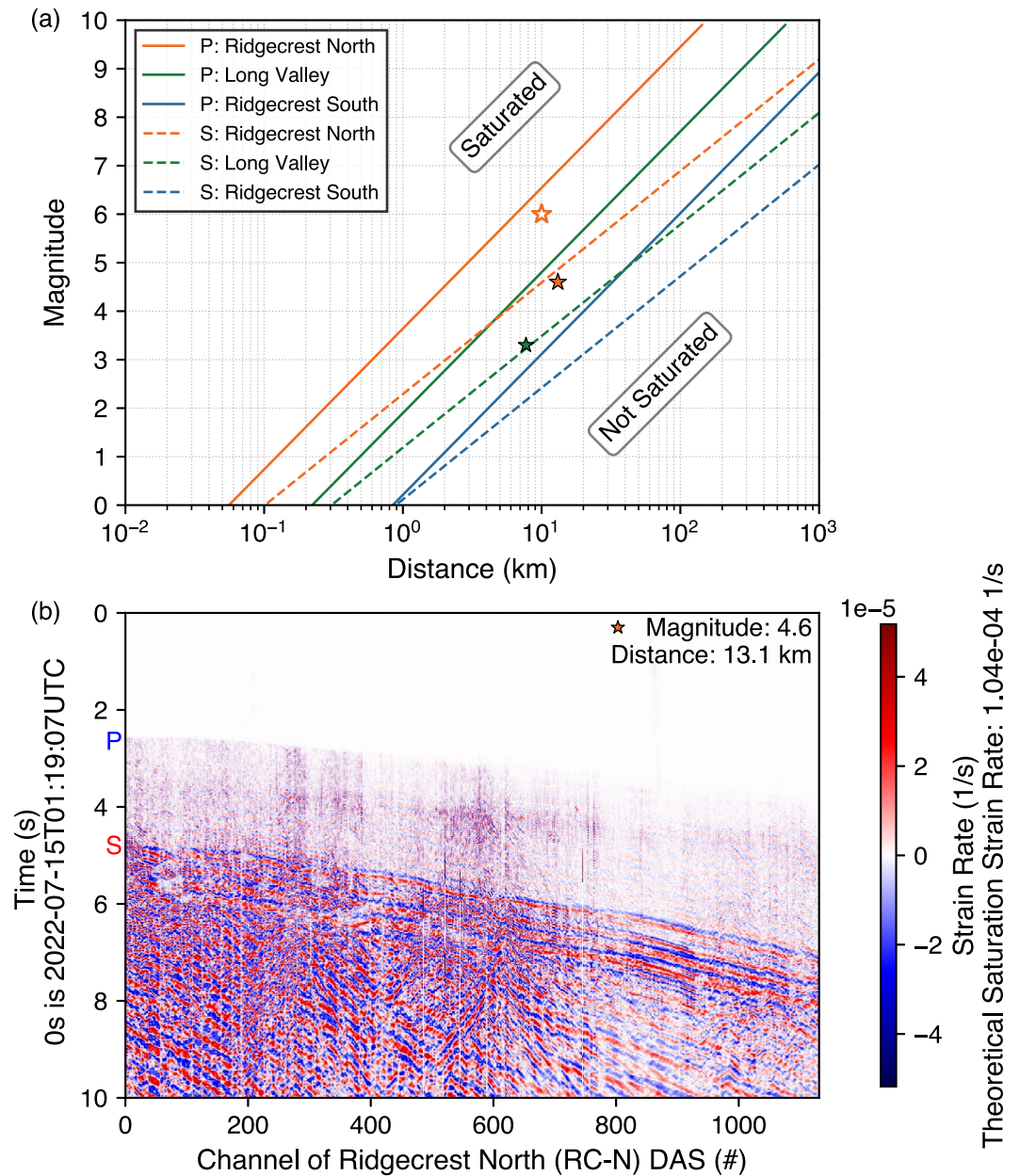


Figure 7. Distributed Acoustic Sensing (DAS) amplitude saturation/clipping threshold of earthquakes and corresponding possible examples for earthquake early warning (EEW). (a) Saturation thresholds for *P* (solid lines) and *S* (dashed lines) waves of the three DAS arrays, with different pulse-repetition-rate to gauge-length ratios. The orange open star represents a possible example of using Ridgecrest North DAS for EEW, as the *P* wave of a magnitude six earthquake is not saturated at a distance of 10 km. The orange solid star represents a real example of a magnitude 4.6 earthquake at a distance of 13.1 km. Its *P* and *S* waves, recorded by the Ridgecrest North DAS, as shown in (b), were not saturated. The green solid star represents a real example of a magnitude 3.3 earthquake at a distance of 7.7 km. Its *P* wave, recorded by the Long Valley DAS, was not saturated. (b) DAS waveforms of the magnitude 4.6 earthquake at a distance of 13.1 km, as marked by the orange solid star in (a). The theoretical saturation strain rate of Ridgecrest North DAS is $1.04 \times 10^{-4} \text{ s}^{-1}$, as shown in Figure 6a.

(Yin, Zhu, et al., 2023) and variations in saturation strain rates across different DAS arrays (Figure 6), we established specific saturation thresholds for *P* and *S* waves for each array (Figure 7a).

Our results suggest that a DAS array with a higher pulse repetition rate and shorter gauge length has a higher saturation threshold (Figure 6). For instance, the real example in Figure 7b demonstrates that the *P* and *S* waves of a magnitude 4.6 earthquake at a distance of 13.1 km are not saturated (the orange solid star in Figure 7a) for the

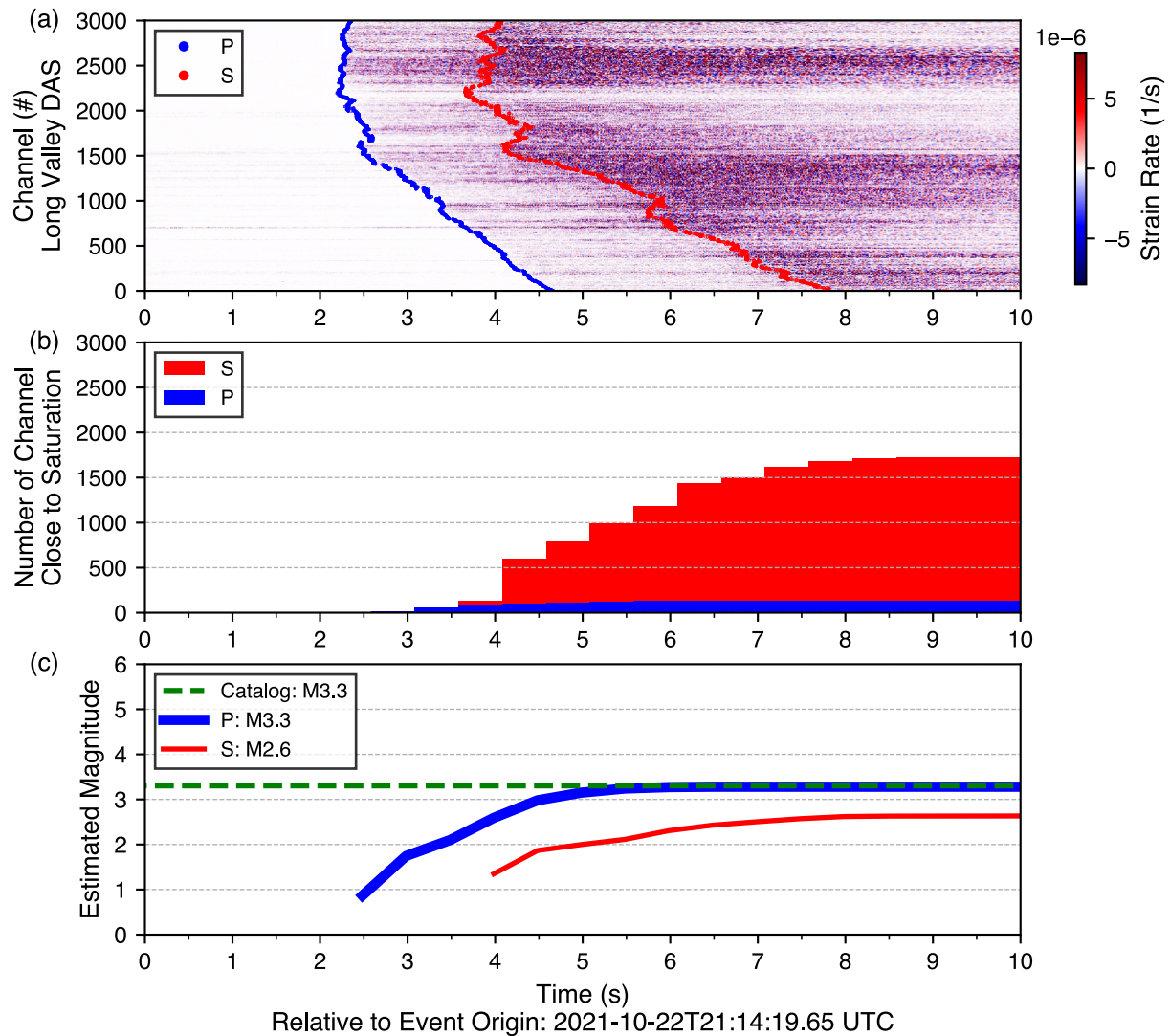


Figure 8. Magnitude estimation of a cataloged magnitude 3.3 earthquake recorded by the Long Valley Distributed Acoustic Sensing (DAS), as marked by the green solid star in Figure 7a. (a) DAS waveforms with phase pickings, (b) Number of channels where *P*- or *S*-wave peak amplitudes are close to saturation (higher than 50% of the theoretical saturation strain rate (Equation 4 and Figure 6a)), (c) The channel-average estimated magnitude based on *P* and *S* waves. The corresponding estimated magnitudes based on the peak *P*- and *S*-wave DAS amplitudes of each channel are shown in Figure S8 in Supporting Information S1.

Ridgecrest North DAS. The orange open star in Figure 7a indicates that the *S* wave of a magnitude six earthquake at a distance of 10 km for the Ridgecrest North DAS could be saturated. However, its *P* wave is not saturated and can be used for EEW. Although a magnitude six earthquake at 10 km for the Ridgecrest North DAS is not available during our experiment, a real example in Figure 8 demonstrates that the *P* wave of a magnitude 3.3 earthquake at a distance of 7.7 km for the LV DAS is not saturated (the green solid star in Figure 7a) and can be used for EEW.

Yin, Soto, et al. (2023) developed a workflow for EEW with DAS that includes event detection, location, and magnitude estimation, and tested it with unsaturated real data. Here, we further develop this workflow by demonstrating how to automatically determine if the *P* or *S* waves of an earthquake are saturated on a DAS array and then estimate the magnitude using only the unsaturated phase(s). First, we apply the pretrained PhaseNet-DAS deep-learning model to automatically pick the *P*- and *S*-wave arrival times (Yin, Soto, et al., 2023; Zhu et al., 2023). Since Yin, Soto, et al. (2023) already demonstrated that phase arrivals can be used for event detection and location, we focus on amplitude saturation and magnitude estimation. Figure 8a shows the DAS waveforms with phase pickings of a magnitude 3.3 earthquake recorded by the LV DAS (green solid star in Figure 7a). As the DAS-measured phase change approaches its limits ($-\pi$ or π), it skips to the next cycle, making the exact point of saturation unclear. To detect saturation, we monitor channels approaching the saturation threshold. For a given

DAS channel, if the observed peak amplitude of the P or S wave is higher than 50% of the theoretical saturation values (Equation 4 and Figure 6a), we label it as a close-to-saturation channel for the corresponding phase (P or S). Figure 8b shows the number of close-to-saturation channels for both P and S waves of this magnitude 3.3 earthquake. The P waves on most channels are not saturated, but the S waves on more than 50% of the channels are saturated. The calculated magnitude based on S waves underestimates the earthquake magnitude (Figure 8c and Figure S8 in Supporting Information S1). However, we can automatically exclude the saturated S phase from the magnitude estimation for EEW if there are a large number of close-to-saturation channels (Figure 8b). The estimated magnitude based solely on unsaturated P waves matches the magnitude in the standard earthquake catalog (Figure 8c and Figure S8 in Supporting Information S1).

We find that the saturation amplitude of DAS is tunable for EEW for earthquakes at a distance of 10 km with magnitudes larger than 6 (Figure 7a). Specifically, our findings suggest that modifications such as increasing the pulse repetition rate and decreasing the gauge length can expand the dynamic range of DAS for EEW (Equation 4 and Figure 6). Although increasing the pulse repetition rate and decreasing the gauge length can expand the dynamic range of DAS, it is important to note that a higher pulse repetition rate may reduce the maximum operable length of the DAS cable (Text S3 in Supporting Information S1), while a shorter gauge length could increase the noise level. Future studies should consider these trade-offs to optimize DAS configurations for specific applications, similar to choosing between conventional broadband stations and strong-motion stations for different applications. Phase unwrapping algorithms may also extend the dynamic range of DAS systems (Ende et al., 2024; Itoh, 1982; Nishiguchi, 2016). Newer DAS interrogators with dual gauge lengths offer another potential solution for optimizing DAS configurations.

Even if part of the waveform (e.g., the S wave) of a larger earthquake is saturated, we demonstrate that we can automatically detect the saturated phase and correctly estimate the earthquake magnitude using only the unsaturated phase(s) (Figure 8). Previous studies have shown that P -wave-based approach can successfully provide EEW services, such as the P -alert system in Taiwan (Y.-M. Wu et al., 2013, 2018). Additionally, even if some DAS channels experience saturation in the S or P waves, the unsaturated channels can still be used if they can be automatically identified by next-generation DAS instruments or future advanced saturation detection algorithm. Since DAS can operate on cables over 100 km long, there will often be a mix of saturated and unsaturated channels. Earthquake early warning algorithms can be developed to utilize only the unsaturated P and S waves, though these may come from farther away from the epicenter, potentially reducing warning times. However, this approach could still be valuable in regions with sparse station spacing in conventional EEW networks. Current EEW algorithms fall into three primary categories: point-source models, finite fault models, and ground-motion models (Allen & Melgar, 2019). The strategy we propose for increasing DAS saturation thresholds and automatically detecting saturated phases and channels should be adaptable to all these types of EEW systems. Our study highlights the significant potential for developing future EEW systems using DAS in practical applications.

6.2. Ambient Noise Seismic Interferometry

In addition to the earthquake source characterization and natural hazard mitigation discussed above, studying Earth's subsurface structure is also a significant focus in geophysics. Recent studies have shown that ambient (e.g., traffic) noise seismic interferometry with DAS is a powerful tool for near-surface structure imaging and monitoring (Ajo-Franklin et al., 2019; Dou et al., 2017; Z. Shen et al., 2024; Spica et al., 2020; Yang, Atterholt, et al., 2022; Yang, Zhan, et al., 2022). In practice, DAS ambient noise cross-correlation (CC) results usually need to be stacked over a time window (ranging from hours to months) to achieve a sufficient SNR for further analysis. The duration of this stacking is particularly important for obtaining robust, high temporal resolution results in subsurface monitoring. For example, if achieving the required SNR for further analysis requires stacking data over weeks, then daily or hourly monitoring becomes unfeasible. Therefore, insights and guidance regarding this CC data quality issue are essential for designing future DAS imaging and monitoring experiments. However, our understanding of this issue remains limited.

To investigate the effect of noise level on the data quality of DAS ambient noise CC results, we analyze the data of Ridgecrest South DAS array. This array was selected because the CC results vary significantly along different segments of the DAS array (Figure 9). We follow the workflow outlined in Bensen et al. (2007) and Yang, Zhan, et al. (2022) to preprocess each one-hour DAS data segment. The preprocessing steps include converting raw data to strain rate, removing mean and linear trends, tapering, band-pass filtering between 0.1 and 20 Hz, removing

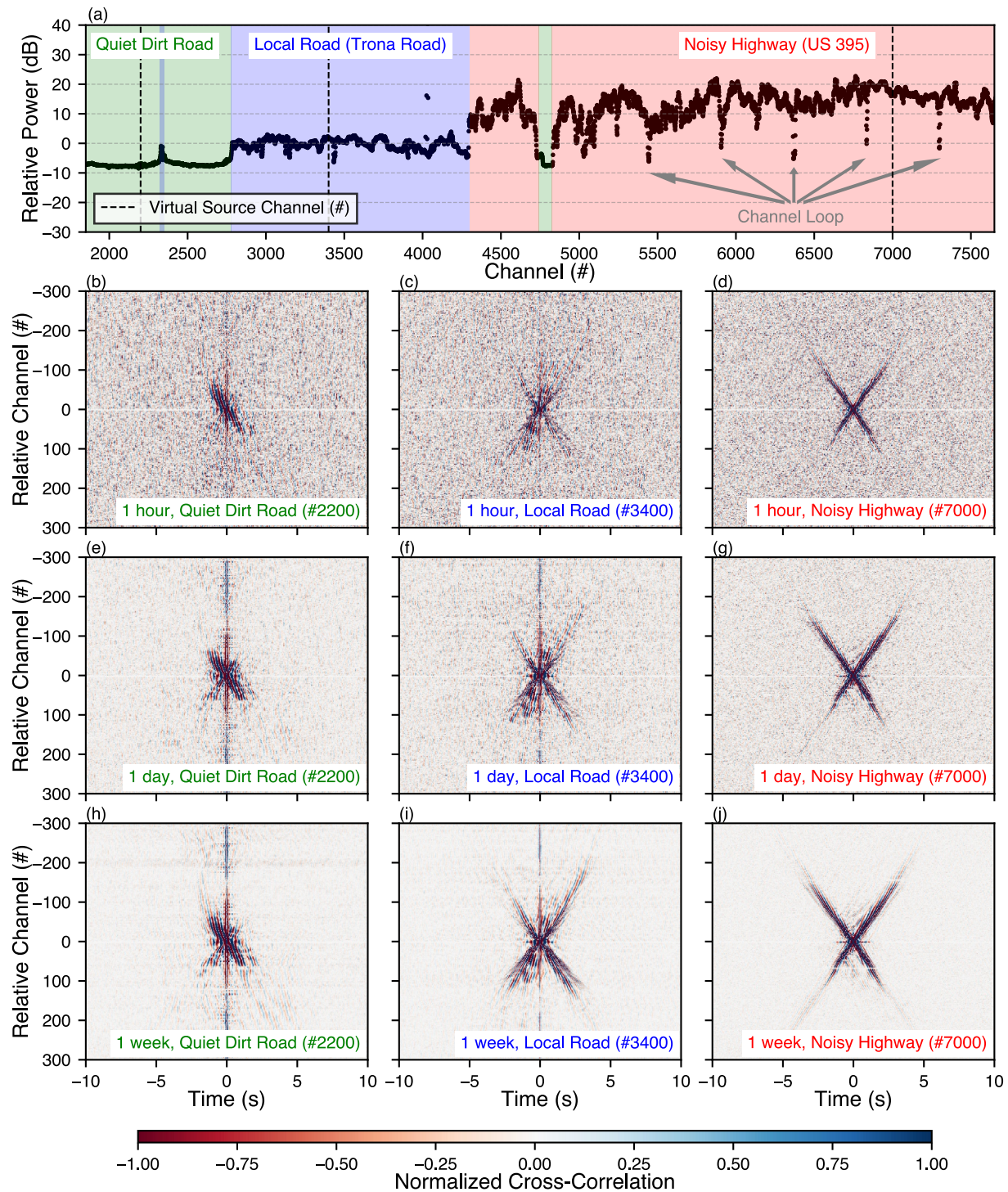


Figure 9. Variations in the data quality of Distributed Acoustic Sensing (DAS) ambient noise interferometry results and corresponding local noise levels across DAS channels. (a) Relative variations in the noise level at 6 Hz across channels along the Ridgecrest South DAS cable, with different shaded colors corresponding to varying traffic levels along the telecommunication cable deployment route (Figure 1d), (b) Example wavefield of DAS ambient noise cross-correlation (CC) using 1 hr continuous data recorded by channels along a quiet dirt road. Channel #2200 (Relative Channel #0), used as the virtual source, is marked by a black vertical dashed line in (a). The CC results are band-pass filtered between 2 and 10 Hz, (c)–(d) Similar to (b), but for the channels along a local road and a noisy highway, respectively, (e)–(g) Similar to (b)–(d), but stacking 1 day CC results, and (h)–(j) Similar to (b)–(d), but stacking 1 week CC results.

common-mode noise, temporal normalization, and spectral whitening. After CC and band-pass filtered between 2 and 10 Hz, the hourly results are stacked over 1 day and 1 week for further comparison and discussion (Figure 9). We also measure the relative variations in the local noise level by calculating PSD across channels along the DAS cable (Figure 9a).

The telecommunication cable of Ridgecrest South DAS array was deployed along roads (Figure 1d). Different sections of the DAS cable run alongside various types of roads with varying traffic levels, including quiet dirt road, local road, and noisy highway (Figure 9a). The PSD results quantitatively show that the noise levels at these three DAS sections vary significantly (Figure 9a). For example, the difference in PSD at 6 Hz between the dirt road and the highway exceeds 20 dB (equivalent to 10 times or one order of magnitude in amplitude). The data quality of ambient noise interferometry CC results at these three DAS sections also varies significantly. The noisy highway section produces the highest quality of CC results, while the quiet dirt road section has the lowest quality of CC results (Figure 9).

Our results suggest that local ambient (traffic) noise levels strongly impact the quality of CC results. To achieve robust results for subsurface imaging with ambient noise interferometry, the stacking duration for the quiet dirt road DAS section must be longer than that for the noisy highway DAS section. Compared to the quiet dirt road DAS section, the noisy highway DAS section enables higher temporal resolution in subsurface monitoring with ambient noise seismic interferometry. These findings are helpful for designing future near-surface imaging and monitoring DAS experiments.

7. Conclusions

In this study, we calibrate three telecommunication fiber-optic DAS arrays with co-located seismometers. We conducted a comprehensive assessment of their performance, focusing on instrument response, noise level, and amplitude saturation. Our findings demonstrate that the average amplitude of earthquake signals recorded by DAS is consistent with seismometer data, showing small fluctuations across different channels and frequencies ranging from 0.01 to 10 Hz. Within this frequency band, the noise floor of DAS is comparable to that of strong-motion stations but notably exceeds that of broadband stations at low frequencies, primarily due to self-instrumental noise. Additionally, our results show that DAS has amplitude saturation/clipping issues resulting from the cycle skipping in the laser phase change measurement. However, these saturation characteristics of DAS can be adjusted by tuning the pulse repetition rate and gauge length, though such adjustments require careful consideration of array length and data quality.

We provide a framework for assessing DAS array performance. Our thorough evaluation underscores the significant potential of DAS amplitude data for enhancing future seismological studies. Despite the limited information on cable-ground coupling for pre-existing telecommunication fiber-optic cables, our study provides essential calibration insights for DAS arrays across various segments of telecommunication fiber-optic network (Figure 1b). This emphasizes the considerable potential for developing robust regional and global DAS networks in the future (Lindsey et al., 2019; Spica et al., 2023; Wuestefeld et al., 2023).

In addition to a comprehensive assessment of DAS array performance, we demonstrate how our new findings enhance our understanding of fiber-optic seismology and its implications for natural hazard mitigation, as well as Earth structure imaging and monitoring. Specifically, our results suggest that with proper settings, DAS can detect *P*-waves from an M6+ earthquake occurring 10 km from the cable without saturation, indicating its viability for EEW. Existing on-land and offshore telecommunication cables offer a cost-effective solution for EEW, particularly in subduction zones where real-time ocean bottom seismic networks are challenging and costly to deploy. Through quantitative comparison and analysis, we also find that local ambient traffic noise levels strongly affect the quality of interferometry results. These findings are valuable for future near-surface imaging and monitoring studies using DAS.

Data Availability Statement

The earthquake catalog and waveforms of co-located seismometers used in this study are publicly available at the Southern California Earthquake Data Center (SCEDC, 2013) and the Northern California Earthquake Data Center (NCEDC, 2014). The example dataset of the 1 month continuous recordings of the Ridgecrest North DAS is

publicly available from the SCEDC Earthquake Data AWS Public Dataset (SCEDC, 2013) at <https://scedc.caltech.edu/data/cloud.html>. The measured peak strain rate amplitude of local earthquakes from multiple DAS arrays is available at Yin (2023). The DAS waveforms of 50 high signal-to-noise ratio local earthquakes are publicly available at Atterholt (2022). The DAS waveforms of five high signal-to-noise ratio teleseismic earthquakes (Table S1 in Supporting Information S1), DAS and station locations, DAS noise level PSD results, and 1D velocity models used in this study are publicly available from Zhai (2024) at <https://doi.org/10.22002/tc6tq-7hz47>. Digital versions of the Peterson New Low-/High-Noise Model (Peterson, 1993) and earthquake models (Clinton & Heaton, 2002) as shown in Figure 5 are publicly available from ObsPy (Krischer et al., 2015) at https://docs.obspy.org/_modules/obspy/signal/spectral_estimation.html. The code for converting DAS data of local earthquakes from strain rate to acceleration using the FDCT (Atterholt et al., 2021; Candès et al., 2006) is publicly available at <https://github.com/atterholt/DAS-unit-conversion>. The seismic data is analyzed using ObsPy (Krischer et al., 2015), publicly available at <https://github.com/obspy/obspy>. Some figures are plotted using PyGMT (Tian et al., 2024), publicly available at <https://github.com/GenericMappingTools/pygmt>. All links were last accessed in January 2025.

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